

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 5 — System Architecture Design

Version: 1.00

Contents

1	Meetings within the project.....	3
2	Unresolved topics.....	4
3	Glossar	5
4	System Architecture Design	6
4.1	High-Level Block Architecture	6
4.2	System Subsystems	8
4.3	Data Flow Normal Operation.....	8
4.4	Data Flow Crash Event Sequence	9
4.5	Interfaces.....	9
4.6	Key Design Principles	9
5	Figure and table listing.....	10
6	List of dependent documents	11
7	Literature	12
8	Version history	13

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	17:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 System Architecture Design

A granular evaluation of the ECU's technical architecture is mandatory to identify the malfunction pathways that compromise safety requirements. In an ASIL D environment, we must account for every potential failure point within the hardware and communication blocks.

4.1 High-Level Block Architecture

The hardware is organised into six primary functional blocks that form the hardware architecture of the Airbag ECU.

Tabelle 2 High-Level Block Architecture

Block	Component (Prototype)	Function
Power Supply	12 V to 5 V / 3.3 V LDO	Provides regulated voltage; capacitor hold-up for post-crash operation
Microcontroller	Arduino Due (ARM Cortex-M3)	Central processing: sensor reading, crash logic, output control, CAN
Sensor Inputs	Analog potentiometers (simulation)	Frontal acceleration (A0), gyroscope/rollover (A1), side pressure (A2)
Digital Input	Push button / switch (D2)	Baby/passenger seat detection signal (INPUT_PULLUP)
Output Drivers	LEDs on D8, D9, D13	Driver airbag, passenger airbag (simulated), warning lamp
CAN Interface	CAN0 (Due built-in) + MCP2551 transceiver	ISO 11898 CAN bus communication; diagnostic frame transmission and reception

Figure 1: Airbag ECU System Architecture Block Diagram — showing all sensors, Satellite Sensor Interface, Sensor SPI Bus, Main CPU, Watchdog, Flash/EEPROM, CAN Transceiver, LIN Transceiver, Safing Engine, Firing SPI Bus, all deployment drivers and actuators.

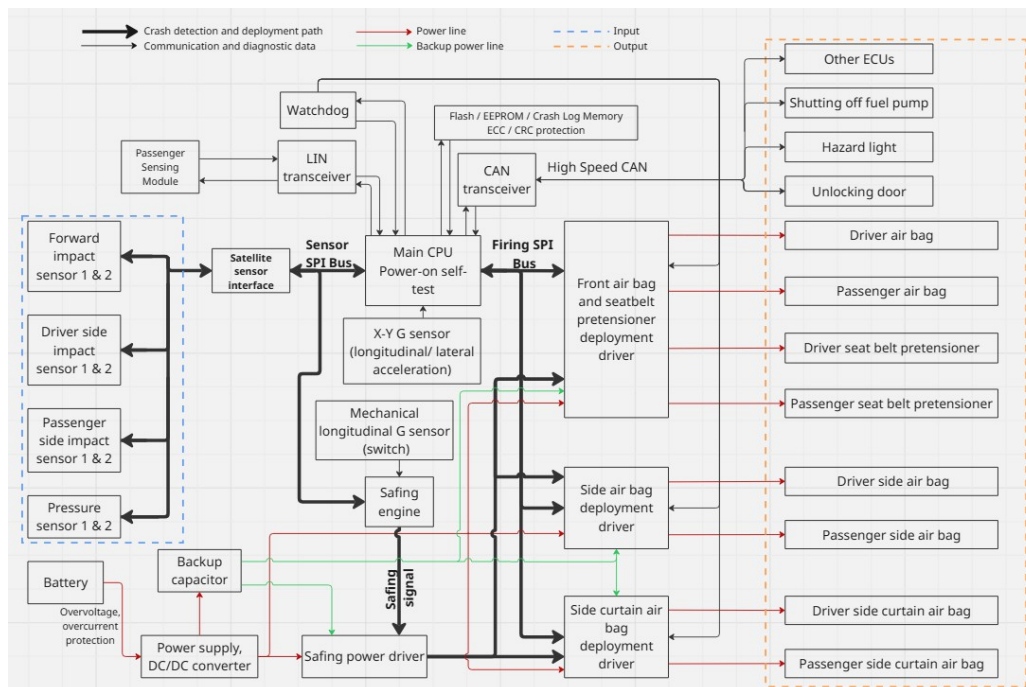


Abbildung 1 Airbag ECU System Architecture Block Diagram (Source: ECU.png, 03_System_Architecture)

Figure 2: Architecture Version 2 Baby Seat Sensing Module and Passenger Sensing Module added to LIN transceiver path.

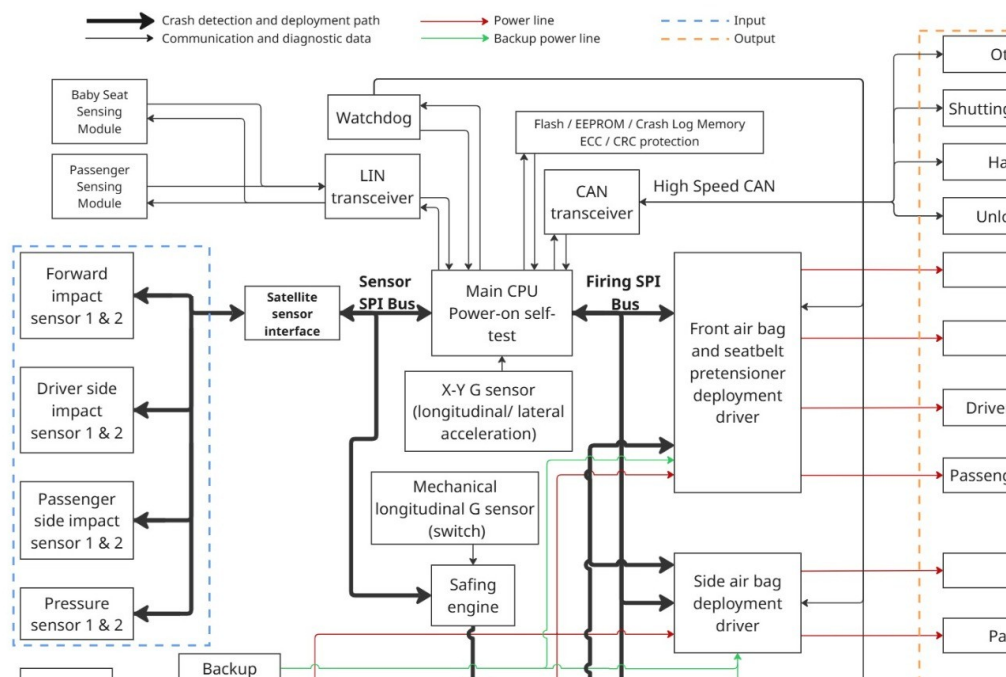


Abbildung 2 Airbag ECU System Architecture V2 (Source: Architecture.jpeg, 03_System_Architecture)

4.2 System Subsystems

Tabelle 3 System Subsystems

Subsystem	Description
Sensor Subsystem	Forward impact sensors (1 and 2), Driver-side impact sensors (1 and 2), Passenger-side impact sensors (1 and 2), Pressure sensors (1 and 2), X-Y G-Sensors (longitudinal and lateral acceleration), Mechanical longitudinal G sensor (hardware switch backup). Sensors communicate via Sensor SPI Bus. Satellite sensors connect through a Satellite Sensor Interface.
Processing Unit (Main CPU)	Performs Power-On Self-Test (POST). Continuously evaluates sensor data. Runs crash detection algorithms. Controls deployment decisions. Connected modules: Watchdog (system monitoring), Flash/EEPROM (crash logs and configuration), CAN and LIN communication interfaces.
Communication Interfaces	CAN Transceiver (High-Speed CAN): communicates with other ECUs, sends crash status and diagnostics. LIN Transceiver: interfaces with passenger sensing module and baby seat sensing module.
Safety Subsystem	Watchdog: monitors CPU health and resets on failure. Safing Engine: independent validation of crash conditions, validates crash condition separately from the main CPU, generates the safing signal, operates in parallel with the main CPU. Mechanical G Sensor: hardware-related redundant trigger.
Deployment Subsystem	Firing SPI Bus: secure communication between CPU and deployment drivers. Deployment Drivers: Front airbag and seatbelt pretensioners driver, Side airbag deployment driver, Side curtain airbag deployment driver.
Actuators (Output)	Driver airbag, Passenger airbag, Driver seatbelt pretensioner, Passenger seatbelt pretensioner, Side airbags (driver and passenger), Curtain airbags (driver and passenger).
Power Supply System	Battery Input (primary power source, overvoltage/overcurrent protection), DC/DC converter (regulates voltage for ECU components), Backup Capacitor (provides energy during power loss e.g. crash disconnection), Safing Power Driver (controls energy delivery to deployment circuits).

4.3 Data Flow Normal Operation

1. Sensors continuously send data via SPI.
2. The main CPU processes acceleration and pressure signals.
3. Crash conditions are evaluated by: (a) Main CPU (software algorithms) and (b) Safing Engine (independent hardware logic).
4. The main CPU communicates with other vehicle ECUs via: (a) CAN (High-speed communication) and (b) LIN (e.g., passenger sensing module, baby seat sensing module).

4.4 Data Flow Crash Event Sequence

Tabelle 4 Data Flow Crash Event Sequence

Step	Action
1	Impact detected by multiple sensors
2	CPU runs crash algorithm
3	Safing engine validates event independently
4	Deployment is allowed only when both conditions are satisfied: Main CPU confirms the crash event AND Safing Engine asserts the safing signal
5	Airbags and pretensioners deploy
6	Crash data stored in Flash memory
7	CAN messages sent to other ECUs (e.g., Fuel cut-off, hazard lights, door power lock)

4.5 Interfaces

Tabelle 5 Interfaces

Interface	Type	Purpose
Sensor SPI Bus	Input	Sensor data acquisition
Firing SPI Bus	Output	Deployment control
CAN	Bidirectional	Vehicle communication
LIN	Bidirectional	Passenger sensing module

4.6 Key Design Principles

Tabelle 6 Key Design Principles

Key Design Principle	Reference
Redundancy	Airbag_ECU_Phase_1, 4.6 Safety Features, Table 6
Deterministic real-time response	Airbag_ECU_Phase_4, 4.1 Timing Requirements, Table 2
Fail-safe operation	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG1
Independence of safety paths	Airbag_ECU_Phase_6, 4.1 Dual-Path Validation, Table 2
Energy availability during crash	Airbag_ECU_Phase_4, 4.1 Timing Requirements, Table 2

5 Figure and table listing

Tabelle 1: Glossar	5
Tabelle 2 High-Level Block Architecture	6
Tabelle 3 System Subsystems	8
Tabelle 4 Data Flow Crash Event Sequence	9
Tabelle 5 Interfaces	9
Tabelle 6 Key Design Principles	9
Abbildung 1 Airbag ECU System Architecture Block Diagram (Source: ECU.png, 03_System_Architecture)	7
Abbildung 2 Airbag ECU System Architecture V2 (Source: Architecture.jpeg, 03_System_Architecture)	7

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf
2	ECU Airbag System Architecture Documentation	V01	07.05.2026	ECU Airbag system architecture documentation_V01.pdf
3	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

No. Reference

- M. Skutek, M. Mekhaïel and G. Wanielik, A precrash system based on radar for automotive applications, IEEE IV2003 Intelligent Vehicles Symposium, Columbus, OH, USA, 2003. doi: 10.1109/IVS.2003.1212879.
- [1]
- [2] RDAIRBAGPSI5 Airbag Reference Platform, Freescale Semiconductor, Inc. Retrieved April 24, 2026, from <https://community.nxp.com>
- [3] A test procedure for airbags Study 2. Retrieved April 24, 2026, from <https://citainsp.org/wp-content/uploads/2016/01/ECS-RSP-Study-2-TP-airbags.pdf>
- [4] audioholic, Technically understanding Airbag systems and SRS, Team-BHP.com, Aug. 12, 2017. <https://www.team-bhp.com/forum/road-safety/189259>
- [5] EEVblog, EEVblog #517 - Car Airbag Controller Teardown, YouTube, Sep. 8, 2013. <https://youtu.be/kcxw850YpVc?t=899>
- [6] Infineon Technologies, Automotive power product brief TLE 7714/7734. https://www.infineon.com/dgdl/TLE7714-34_PB_final.pdf
- [7] element14 Community, Freescale's automotive DSI airbag system, Jun. 20, 2012. <https://community.element14.com>
- [8] Guide to Airbag Control Unit ACU - Enhancing Automotive Safety, embien. <https://www.embien.com/automotive-insights/guide-to-airbag-control-unit-acu-enhancing-automotive-safety>

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_5	09.06.2026	v1.00 Initial version. System Architecture Design compiled from Architecture V2 Sections 3-6, 9 and HW Concept Design Report V02 Section 3.